

DIP SARKAR

Senior Engineer, Rynco Engineering Ltd., Dhaka, Bangladesh

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RESEARCH INTERESTS

Optical Wireless Communication, Smart Grid Energy, Computer Network, AI/ML, Health AI

EDUCATION

Ahsanullah University of Science and Technology, Dhaka, Bangladesh July 2021 — Dec 2025
Master of Science in Electrical and Electronic Engineering CGPA: 3.08/4.00
Thesis Title: Performance Analysis of Underwater Optical Wireless Communication System under Clear Oceanic Environment

American International University - Bangladesh, Dhaka, Bangladesh Jan 2017 — Apr 2021
Bachelor of Science: Electrical and Electronic Engineering CGPA: 3.50/4.00
Project Title: Different Type of Blade Comparison Incase for Wind Turbine

RESEARCH EXPERIENCE

Elite Research Lab LLC Dhaka, Bangladesh
Research Intern July 2026 — Present

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PUBLICATIONS

Journal paper

Published:

- Sarkar, D., and Barua, B. (2025). Performance analysis of underwater optical wireless communication system under clear oceanic environment. *Journal of Optical Communications*. <https://doi.org/10.1515/joc-2025-0106>
- Hamim, S. J., Hossain, F. S., Hasan, M. M., Sarkar, D., and Mainuddin, M. (2022). A time of use tariff based demand side management algorithm for residential consumers. In *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI)* (pp. 1–6). IEEE. <https://doi.org/10.1109/STI56238.2022.10103303>

SELECTED COURSES

Master's Courses

- Optical Networks
- Digital Satellite Communications
- Smart Grid Design and Application
- Computer Methods in Power System Analysis
- Renewable Energy Resources and Technologies
- Compound Semiconductor Devices

Bachelor's Courses

- Telecommunications Engineering
- Optoelectronic Devices
- Modern Control Systems
- Power Systems Analysis
- Digital Signal Processing
- Industrial Electronics and Drives
- Digital Logic and Circuits
- Electromagnetics Fields and Waves
- Analog Electronics
- Signals and Linear Systems
- Electrical Power Transmission and Distribution
- Electronic Devices
- Renewable Energy Technology
- Numerical Methods for Science and Engineering

ENGLISH TEST

IELTS (Academic): 7.5 (overall score)

Listening: XX — Reading: XX

Speaking: XX — Writing: XX

Test date: August 2026

SKILLS

- **Programming:** Python, C, SQL, MATLAB
- **AI/ML (Frameworks):** PyTorch, TensorFlow, Keras
- **AI/ML (Libraries):** Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- **Tools:** Google Colab, NotebookLM, Git, LaTeX, Google Cloud Platform

REFERENCES

Prof. Dr. Bobby Barua

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Scholar Profiles: AUST Profile — Google Scholar — ORCID

Dr. Taskin Jamal

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Prof. Dr. Md. Saniat Rahman Zishan

professor and Director, EEE, AIUB, Dhaka, Bangladesh

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Scholar Profiles: AIUB Profile — Google Scholar — ResearchGate